

Reviewer Pack — Additive Energy Gap Between P and AC⁰ Functions

Entry 735ee8b65843 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-07 19:41:50 UTC

Additive Energy Gap Between P and AC⁰ Functions

Entry ID: 735ee8b65843 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-07 19:41:50 UTC

1. Conjecture

Field A (mathematical branch): Additive Combinatorics **Field B** (complexity object): Boolean Circuit Complexity

Statement:

For any explicit Boolean function $f \in P$ with n variables, its additive energy $E(f) = |\{(x,y,z,w) \in \{0,1\}^n : f(x) + f(y) = f(z) + f(w)\}|$ satisfies $E(f) \leq O(n^2)$. Conversely, for any AC⁰ circuit computing f , $E(f) \geq \Omega(2^n/n^2)$.

Rationale (proposer's reasoning):

Additive energy captures the structured regularity of a function's truth table. AC⁰ circuits, with their bounded-depth structure, inherently exhibit higher additive energy due to correlation with low-degree polynomials, while explicit P-functions (e.g., parity, modular counting) have pseudorandom truth tables with sparse additive structure.

Taxonomy category: ACC_SIPSER (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): e600c0abacde281b

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
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4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - additive energy Boolean circuits AC^0 - explicit Boolean functions P vs AC^0 additive energy - additive energy P AC^0 bounds

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, $\leq 90s$ wall time per cycle **Execution:** rc=124, elapsed=240.1s

5.1 Generated Python source

```
import random

def run_trial(seed: int) -> dict:
    n = 30
    random.seed(seed)

    def add(x, y):
        return x + y

    def bitwise_and(x, y):
        return x & y

    def bitwise_or(x, y):
        return x | y

    def parity(f):
        return f % 2

    def ac0_circuit(f):
        if n == 5:
            return (f[0] and f[1]) or (f[2] and f[3])
        elif n == 10:
            return ((f[0] and f[1]) or (f[2] and f[3])) and ((f[4] and f[5]) or (f[6] and f[7]))
        elif n == 15:
            return (((f[0] and f[1]) or (f[2] and f[3])) and ((f[4] and f[5]) or (f[6] and f[7])))
                (((f[8] and f[9]) or (f[10] and f[11])) and ((f[12] and f[13]) or (f[14] and f[15])))
        elif n == 20:
            return (((((f[0] and f[1]) or (f[2] and f[3])) and ((f[4] and f[5]) or (f[6] and f[7])))
                (((f[8] and f[9]) or (f[10] and f[11])) and ((f[12] and f[13]) or (f[14] and f[15])))
                (((f[16] and f[17]) or (f[18] and f[19])) and ((f[20] and f[21]) or (f[22] and f[23]))
                (((f[24] and f[25]) or (f[26] and f[27])) and ((f[28] and f[29]) or (f[30] and f[31]))
        elif n == 30:
            return ((((((f[0] and f[1]) or (f[2] and f[3])) and ((f[4] and f[5]) or (f[6] and f[7])))
                (((f[8] and f[9]) or (f[10] and f[11])) and ((f[12] and f[13]) or (f[14] and f[15])))
                (((f[16] and f[17]) or (f[18] and f[19])) and ((f[20] and f[21]) or (f[22] and f[23]))
                (((f[24] and f[25]) or (f[26] and f[27])) and ((f[28] and f[29]) or (f[30] and f[31]))
                (((f[32] and f[33]) or (f[34] and f[35])) and ((f[36] and f[37]) or (f[38] and f[39]))
                (((f[40] and f[41]) or (f[42] and f[43])) and ((f[44] and f[45]) or (f[46] and f[47]))
```

```

((((f[48] and f[49]) or (f[50] and f[51])) and ((f[52] and f[53]) or (f[54] and
(((f[56] and f[57]) or (f[58] and f[59])) and ((f[60] and f[61]) or (f[62] and
((((f[64] and f[65]) or (f[66] and f[67])) and ((f[68] and f[69]) or (f[70] and
(((f[72] and f[73]) or (f[74] and f[75])) and ((f[76] and f[77]) or (f[78] and
((((f[80] and f[81]) or (f[82] and f[83])) and ((f[84] and f[85]) or (f[86] and
(((f[88] and f[89]) or (f[90] and f[91])) and ((f[92] and f[93]) or (f[94] and
((((f[96] and f[97]) or (f[98] and f[99])) and ((f[100] and f[101]) or (f[102] a
(((f[104] and f[105]) or (f[106] and f[107])) and ((f[108] and f[109]) or (f[11
((((f[112] and f[113]) or (f[114] and f[115])) and ((f[116] and f[117]) or (f[11
(((f[120] and f[121]) or (f[122] and f[123])) and ((f[124] and f[125]) or (f[12
((((f[128] and f[129]) or (f[130] and f[131])) and ((f[132] and f[133]) or (f[13
(((f[136] and f[137]) or (f[138] and f[139])) and ((f[140] and f[141]) or (f[14
((((f[144] and f[145]) or (f[146] and f[147])) and ((f[148] and f[149]) or (f[15
(((f[152] and f[153]) or (f[154] and f[155])) and ((f[156] and f[157]) or (f[15
((((f[160] and f[161]) or (f[162] and f[163])) and ((f[164] and f[165]) or (f[16
(((f[168] and f[169]) or (f[170] and f[171])) and ((f[172] and f[173]) or (f[17
((((f[176] and f[177]) or (f[178] and f[179])) and ((f[180] and f[181]) or (f[18
(((f[184] and f[185]) or (f[186] and f[187])) and ((f[188] and f[189]) or (f[19
((((f[192] and f[193]) or (f[194] and f[195])) and ((f[196] and f[197]) or (f[19
(((f[200] and f[201]) or (f[202] and f[203])) and ((f[204] and f[205]) or (f[20
((((f[208] and f[209]) or (f[210] and f[211])) and ((f[212] and f[213]) or (f[21
(((f[216] and f[217]) or (f[218] and f[219])) and ((f[220] and f[221]) or (f[22
((((f[224] and f[225]) or (f[226] and f[227])) and ((f[228] and f[229]) or (f[23
(((f[232] and f[233]) or (f[234] and f[235])) and ((f[236] and f[237]) or (f[23
((((f[240] and f[241]) or (f[242] and f[243])) and ((f[244] and f[245]) or (f[24
(((f[248] and f[249]) or (f[250] and f[251])) and ((f[252] and f[253]) or (f[25
((((f[256] and f[257]) or (f[258] and f[259])) and ((f[260] and f[261]) or (f[26
(((f[264] and f[265]) or (f[266] and f[267])) and ((f[268] and f[269]) or (f[27
((((f[272] and f[273]) or (f[274] and f[275])) and ((f[276] and f[277]) or (f[27
(((f[280] and f[281]) or (f[282] and f[283])) and ((f[284] and f[285]) or (f[28
((((f[288] and f[289]) or (f[290] and f[291])) and ((f[292] and f[293]) or (f[29
(((f[296] and f[297]) or (f[298] and f[299])) and ((f[300] and f[301]) or (f[30
((((f[304] and f[305]) or (f[306] and f[307])) and ((f[308] and f[309]) or (f[31
(((f[312] and f[313]) or (f[314] and f[315])) and ((f[316] and f[317]) or (f[31
((((f[320] and f[321]) or (f[322] and f[323])) and ((f[324] and f[325]) or (f[32
(((f[328] and f[329]) or (f[330] and f[331])) and ((f[332] and f[333]) or (f[33
((((f[336] and f[337]) or (f[338] and f[339])) and ((f[340] and f[341]) or (f[34
(((f[344] and f[345]) or (f[346] and f[347])) and ((f[348] and f[349]) or (f[35
((((f[352] and f[353]) or (f[354] and f[355])) and ((f[356] and f[357]) or (f[35
(((f[360] and f[361]) or (f[362] and f[363])) and ((f[364] and f[365]) or (f[36
((((f[368] and f[369]) or (f[370] and f[371])) and ((f[372] and f[373]) or (f[37
(((f[376] and f[377]) or (f[378] and f[379])) and ((f[380] and f[381]) or (f[38
((((f[384] and f[385]) or (f[386] and f[387])) and ((f[388] and f[389]) or (f[39
(((f[392] and f[393]) or (f[394] and f[395])) and ((f[396] and f[397]) or (f[39
((((f[400] and f[401]) or (f[402] and f[403])) and ((f[404] and f[405]) or (f[40
(((f[408] and f[409]) or (f[410] and f[411])) and ((f[412] and f[413]) or (f[41
((((f[416] and f[417]) or (f[418] and f[419])) and ((f[420] and f[421]) or (f[42
(((f[424] and f[425]) or (f[426] and f[427])) and ((f[428] and f[429]) or (f[43
((((f[432] and f[433]) or (f[434] and f[435])) and ((f[436] and f[437]) or (f[43
(((f[440] and f[441]) or (f[442] and f[443])) and ((f[444] and f[445]) or (f[44
((((f[448] and f[449]) or (f[450] and f[451])) and ((f[452] and f[453]) or (f[45

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase time limit and re-run test with stricter resource constraints

11. Audit log (LLM calls)

Total LLM calls: 8

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	104703
2	propose	ollama_remote	qwen3:8b	0	0	61848
3	preregistration	ollama_remote	qwen3:8b	0	0	19913
4	novelty	ollama_remote	qwen3:8b	0	0	15668
5	novelty	ollama_remote	qwen3:8b	0	0	10323
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	43946
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	45415
8	judge	ollama_remote	qwen3:8b	0	0	10249

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 312064 ms total latency. Provider mix: {'ollama_remote': 8}

(full prompt+response transcripts available in `research/audit/735ee8b65843.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/735ee8b65843.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/735ee8b65843.tar.gz` (if generated)